9.2 Combining Two Functions: Sums and Differences

A Definitions

The *sum* and *difference* of two functions are defined by

$$(f+g)(x) = f(x) + g(x)$$
$$(f-g)(x) = f(x) - g(x)$$

Ex 1. Let $f(x) = x^2 - 1$ and $g(x) = 1 + \sqrt{x+1}$. Find

a) $(f+g)(0) = f(0) + g(0)$

$= -1 + 2 = 1$

b) $(f-g)(3) = f(3) - g(3)$

$= 8 - 3 = 5$

c) $(f+g)(-2) = f(-2) + g(-2)$

$= 3 + 1 + \sqrt{-1}$ ∴ undefined

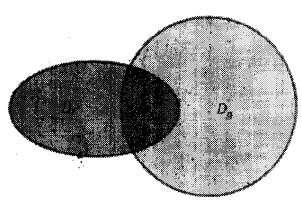
d) $(f+g)(x) = x^2 - \cancel{1} + \cancel{1} + \sqrt{x+1}$

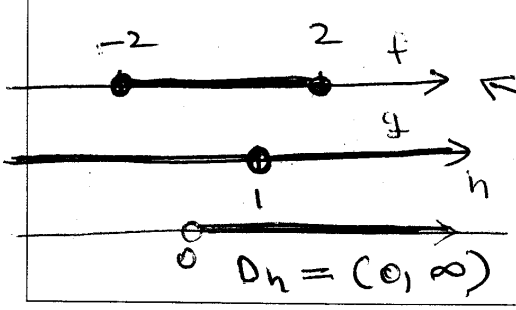
$= x^2 + \sqrt{x+1}$

B Domain of Sum and Difference of Two Functions

The *domain* of the sum or difference of two functions is the *intersection* of their domains.

$$D_{f \pm g} = D_f \cap D_g$$





$D_h = (0, \infty)$

Ex 2. For each case, find the domain of the sum of the given functions.

a) $f(x) = e^x$; $g(x) = 1/x^2$

$D_f = \mathbb{R}$; $D_g = \{x \neq 0\}$

$D_{f+g} = \{x \in \mathbb{R} \mid x \neq 0\}$

b) $f(x) = \sqrt{2-x}$; $g(x) = \log(x+1)$

$D_f = (-\infty, 2]$; $D_g = (-1, \infty)$

$D_{f+g} = (-1, 2]$

c) $f(x) = 2^{x-2}$; $g(x) = x^3 + 1$

$D_f = \mathbb{R}$; $D_g = \mathbb{R}$

$D_{f+g} = \mathbb{R}$

d) $f(x) = \sqrt{4-x^2}$; $g(x) = \frac{1}{x-1}$; $h(x) = \log x$

$D_f = [-2, 2]$; $D_g = (-\infty, 1) \cup (1, \infty)$

$\therefore D_{f+g+h} = (0, 1) \cup (1, 2)$

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C Point by Point

Evaluate $f \pm g$ at every possible number x.

$f(1)=0$ | $g(0)=1$
$f(0)=-1$ | $g(2)=-1$
$f(-1)=2$ | $g(1)=3$

Ex 3. Given

$f=\{(1,0),(0,-1),(-1,2)\}$
$g=\{(0,1),(2,-1),(1,3)\}$

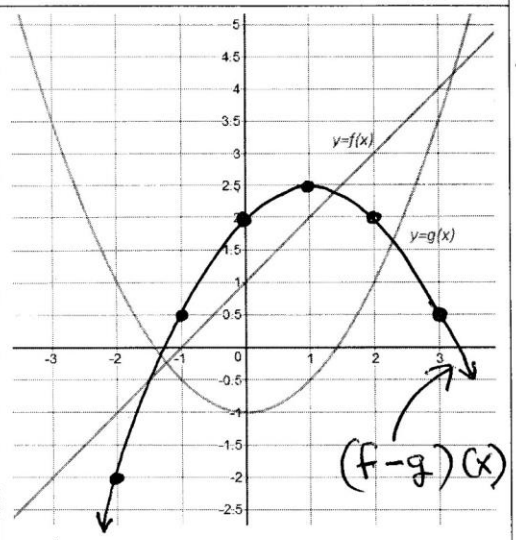
Find $f+g$.

$\therefore f+g=\{(1,3),(0,0)\}$

$(f+g)(1)=f(1)+g(1)=0+3=3$
$(f+g)(0)=f(0)+g(0)=-1+1=0$

Ex 4. The functions f and g are given by their graphs on the right figure. Graph the function $f-g$.

$(f-g)(0)=1-(-1)=2$
$(f-g)(1)=2-(-0.5)=2.5$
$(f-g)(2)=3-1=2$
$(f-g)(3)=4-(3.5)=0.5$
$(f-g)(-1)=0-(-0.5)=0.5$
$(f-g)(-2)=-1-1=-2$



Ex 5. Complete the following table. Justify your reasoning.

	$f+g$	even	odd
g	even	even	neither
	odd	neither	odd

Ex 6. For each case, justify your answer.

a) Is the sum of two polynomial functions a polynomial function?

yes

b) Is the difference of two rational functions a rational function?

yes

~~c)~~ Is the sum of two sine functions a sine function?

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yes if $T_1=T_2$

~~d)~~ Is the sum of two periodic functions a periodic function?

SKIP

yes if $\frac{T_1}{T_2}$ is rational number

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Ex 7. Write the following functions as a sum or a difference of two other functions.

a) $f(x)=\frac{1}{x^2-9}$

$=\frac{1}{6(x-3)}-\frac{1}{6(x+3)}$

b) $f(x)=\log\frac{x}{x+1}$

$=\log x-\log(x+1)$

c) $f(x)=\sin(x-\pi/4)$

$=(\sin x)\frac{\sqrt{2}}{2}-(\cos x)\frac{\sqrt{2}}{2}$

Ex 8. Is the sum of two increasing functions, increasing, decreasing or neither? Give examples to justify your answer.

is increasing

f is increasing when $x_1<x_2$

$f(x_1)<f(x_2)$

g is increasing $g(x_1)<g(x_2)$

$f(x_1)+g(x_1)<f(x_2)+g(x_2)$

$(f+g)(x_1)<(f+g)(x_2)$

SKIP

Ex 9. If $Z_f=\{1,2,3\}$ is the set of all zeros of the function f and $Z_g=\{0,1,2\}$ is the set of all zeros of the function g, what could you say about the set of all zeros of the function $f+g$? Explain your reasoning.

$Z_{f+g}=Z_f\cap Z_g$

SKIP

$Z_{f+g}=\{1,2\}$

$f(x_{int})=0$ $g(x_{int})=0$

$(f+g)(x_{int})=0$

Ex 10. Let $f(x)=\sin x$ and $g(x)=\cos x$. Write $f+g$ and $f-g$ as:

a) a single sine function

$\sin x-\cos x$

$=\frac{1}{\sqrt{2}}\left(\frac{\sqrt{2}}{2}\sin x-\frac{\sqrt{2}}{2}\cos x\right)$

$=\frac{1}{\sqrt{2}}\sin\left(x-\frac{\pi}{4}\right)$

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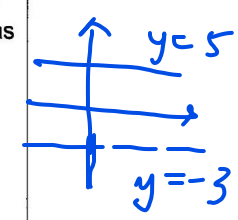
b) a single cosine function

$=\frac{1}{2}\cos\left(x+\frac{\pi}{4}\right)$

Ex 11. The rational function $y=f(x)$ has a horizontal asymptote $y=5$ and the rational function $y=g(x)$ has a horizontal asymptote $y=-3$. What could you say about the horizontal asymptote of the functions:



a) $f+g$

HA $y=2$

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b) $f-g$

$f-g$ HA $y=8$

Reading: Nelson Textbook, Pages 521-528
Homework: Nelson Textbook, Page 528 #3, 4, 5, 7, 10, 13, 16

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